

Project LEE Version 12.0

Full Implementation-Verification Audit

Audit mode: read-only; no repository changes were made.
Repository: main, commit d103c43. Audit date: August 20, 2026.

Executive conclusion

Project LEE has a substantial implementation surface. Most Version 12 engine names have corresponding source modules, database tables, routes, and some UI. Under the requested verification standard, however, no task qualifies as FULLY IMPLEMENTED. The repository is a broad, partially wired operating-intelligence platform with real persistence and several real governance boundaries, but without the integration tests, complete replay model, true restore path, external CIL runtime, or end-to-end verification required to claim Version 12 completion.

Status summary

FULLY IMPLEMENTED: 0 (0%). PARTIAL: 67 (97.1%). SCAFFOLD ONLY: 1 (1.4%). IMPLEMENTED DIFFERENTLY THAN SPECIFIED: 1 (1.4%). No named module is wholly missing, but several required behaviors are absent.

Task classification matrix

#	Task Status Evidence / limitation
1	Foundation & Core Schema PARTIAL lib/db/src/schema; lib/foundation-events.ts. Real schema/event append, but not full event sourcing and no comprehensive tests.
2	Console Web App PARTIAL artifacts/lee-console/src/App.tsx and page components. Live API calls exist; route coverage and governed writes are incomplete.
3	Understanding Pipeline PARTIAL lib/understanding-pipeline.ts; understanding schema/routes. Auditable sources/chunks/runs exist; quality and failure tests absent.
4	Brief Engine PARTIAL core brief table; routes/time.ts; scheduler. Persistence/generation exist; complete CIL path unproven.
5	Model Router & Context PARTIAL model-router.ts; context-engine.ts; context-economy.ts. CIL unavailable; fallback exists; universal ordered flow unproven.
6	Connector Engine PARTIAL connector-engine.ts; connectors.ts; provider-abstraction.ts. Provider-neutral data exists; live sync/retry tests absent.
7	Android App PARTIAL lee-android app, LeeContext.tsx, storage.ts, api.ts. Local-first behavior exists; device restart/reconciliation tests absent.
8	Cost Engine IMPLEMENTED DIFFERENTLY THAN SPECIFIED Superseded by System Economics Task 69, as Version 12 specifies.
9	Backup, Migration & Brain Versioning PARTIAL routes/backups.ts; backup and brain-version schemas. Backup/checksum metadata exists; restore does not rebuild state.
10	Orchestration & Scheduler PARTIAL orchestration.ts; scheduler.ts; durable job schemas/routes. Restart, handler, and retry coverage incomplete.
11	Governance Engine PARTIAL governance-engine.ts; governance routes/schema. Risk/evidence rules exist; every consequential path is not proven CerbaSeal-backed.
12	Memory Architecture PARTIAL memory-architecture.ts; memory schema/routes. Tiers exist; promotion/contradiction lifecycle not comprehensively tested.
13	Intelligence Graph PARTIAL impact.ts; projector.ts; graph schema/routes. Graph/traversal exists; projection replay is narrow.
14	Identity & Relationship Engine PARTIAL identity.ts; founder-identity.ts; relationship schemas/routes. Data persists; ordering enforcement not global.
15	Curiosity Engine PARTIAL curiosity.ts and routes/schema. Candidate generation exists; quality, deduplication, outcomes lack tests.
16	Strategy Engine PARTIAL strategy.ts and routes/schema. Records exist; full strategy-to-execution loop incomplete.
17	Reflection Engine PARTIAL experience.ts; learning.ts; learning routes. Reflection code exists; durable event-to-reflection proof incomplete.
18	Operating Modes PARTIAL recovery-modes.ts; recovery schema/routes. Recovery exists; ordinary mode enforcement limited.

19 Constitution Engine PARTIAL constitution.ts and routes/schema. Provisions persist; universal enforcement absent.

20 Confidence Propagation PARTIAL confidence.ts; operational-confidence.ts. Computation exists; invalidation/source propagation lacks tests.

21 Fact/Interpretation Separation PARTIAL foundation schema; ledgers routes; understanding-pipeline.ts. Separate ledgers real; all write paths untested.

22 Why Chain & Provenance PARTIAL why-chain.ts; provenance.ts; provenance routes. Retrieval exists; provenance-required rejection incomplete.

23 Assumption Ledger PARTIAL assumptions.ts; foundation schema/routes. Validation/invalidation/expiry exist; affected conclusions not fully propagated.

24 Decision Impact Graph PARTIAL impact.ts; graph schema/routes. Directional traversal exists; causal integrity not tested.

25 Digital Twin Timeline PARTIAL timeline.ts and routes/schema. Projection exists; deterministic Event Log rebuild unproven.

26 Query Engine PARTIAL query-engine.ts; query route; context-engine.ts. Central module exists, but many engines query DB directly.

27 Explanation Engine PARTIAL explanation.ts; explanation routes; Why Chain. Sources exist; calibration/feedback tests absent.

28 Semantic Index PARTIAL semantic-index.ts; semantic routes/schema. Local index exists; local-only/rebuild behavior untested.

29 Policy Engine PARTIAL policy.ts; policy routes/schema. Versioned policies exist; enforcement not universal.

30 Resource Engine PARTIAL resource.ts; resources routes/schema. State/reservations exist; orchestration gating unproven.

31 Intent Engine PARTIAL intent.ts; intent routes/schema. Classification/persistence exist; downstream reuse not universal.

32 State Engine PARTIAL state.ts; state routes/schema. Transitions exist; event reconstruction/invariants lack tests.

33 Contracts & Capability Registry PARTIAL internal-contracts.ts; capability-registry.ts; service/provider schemas. Registration exists; self-test does not execute contracts.

34 Context Economy PARTIAL context-economy.ts; context-engine.ts. Ranking/budget structures exist; universal adoption/restart proof absent.

35 Domain Events PARTIAL domain-events.ts; foundation-events.ts. Catalog/validation exist; delivery is in-process only.

36 Lifecycle/Dependencies/Recovery PARTIAL engine-lifecycle.ts; capability-registry.ts; recovery-modes.ts. Metadata exists; operational dependency validation limited.

37 Self-Test Framework SCAFFOLD ONLY self-test.ts and routes/schema. Suites mostly inspect metadata, synthetic events, or assumptions.

38 Recovery Modes PARTIAL recovery-modes.ts and routes/schema. Boot state persists; recovery execution unproven.

39 Data Ownership PARTIAL ownership.ts and ownership routes/schema. Metadata exists; enforcement incomplete.

40 Knowledge Aging PARTIAL knowledge-aging.ts and aging routes/schema. Aging/jobs exist; retrieval gating lacks integration tests.

41 System Manifest PARTIAL system-manifest.ts and manifest route/schema. Manifest exists; intuitive /api/manifest returned 404 in live check.

42 World State PARTIAL world-state.ts and routes/schema. Signals/snapshots exist; external refresh not fully exercised.

43 Operational Memory PARTIAL operational-memory.ts and routes/schema. Signals/patterns exist; long-term evidence quality unproven.

44 Initiative Engine PARTIAL initiative.ts and routes/schema. Generate/acknowledge/dismiss exist; prioritization reliability lacks tests.

45 Operational Intelligence PARTIAL operational-intelligence.ts and routes/schema. Ranking/scheduler exist; live evidence-driven priority not demonstrated.

46 Provider Abstraction PARTIAL provider-abstraction.ts; connectors.ts; provider registry. Adapter structures exist; live sync/boundary enforcement incomplete.

47 Project Bootstrap PARTIAL project-bootstrap.ts; bootstrap routes/schema. Static evidence-first code exists; full repository/security tests absent.

48 CIL + CerbaSeal Services PARTIAL services/internal-services.ts; model-router.ts; integration specs. Signed contracts exist; CIL unavailable and CerbaSeal health returned HTTP 404.

49 Executive Loop PARTIAL executive-loop.ts; routes/schema; scheduler. Persisted ticks exist; phase/restart behavior lacks tests.

50 Operational Confidence PARTIAL operational-confidence.ts and routes/schema. Computation exists; output propagation not fully tested.

51 Project Momentum PARTIAL project-momentum.ts and routes/schema. Scores/history exist; causal evidence not tested.

52 Opportunity Engine PARTIAL opportunity.ts and routes/schema. Detection/persistence exist;

deduplication/outcomes incomplete.

53 Capacity Awareness PARTIAL operational-capacity.ts and routes/schema. Capacity computed; orchestration enforcement unproven.

54 Strategic Anchors PARTIAL strategic-anchors.ts; anchor schema/routes. Persistence/retrieval exist; contradiction protection incomplete.

55 Portfolio Intelligence PARTIAL portfolio-intelligence.ts and routes/schema. Summaries exist; multi-project prioritization untested.

56 Identity Engine PARTIAL identity.ts; founder-identity.ts; identity routes/schema. Profile exists; Identity-before-Constitution not universal.

57 Executive Objectives PARTIAL executive-objectives.ts and routes/schema. Objectives/evidence/status exist; alignment proof incomplete.

58 Organizational Memory PARTIAL organizational-memory.ts and routes/schema. Profiles/context exist; institutional learning partial.

59 Decision Memory PARTIAL decision-memory.ts and routes/schema. Heuristics/records exist; repeated-outcome validation incomplete.

60 Simulation Engine PARTIAL strategy.ts; time-machine.ts; simulation schema/routes. Structures/routes exist; real execution/accuracy unproven.

61 Time Machine PARTIAL time-machine.ts and routes/schema. Historical querying exists; deterministic reconstruction unproven.

62 Uncertainty Tracking PARTIAL uncertainty.ts; routes/schema; Android UncertaintyNotice.tsx. Persistence/visibility exist; complete propagation lacks tests.

63 Resource Allocation PARTIAL resource-allocation.ts and routes/schema. Allocations/overrides/expiry exist; regression coverage absent.

64 Execution Readiness PARTIAL execution-readiness.ts and routes/schema. Live recomputation works; sample had empty evidence and limited tests.

65 Portfolio Dependency Graph PARTIAL portfolio-dependency.ts; routes; graph schema. Synthesis/blast radius work; persistence/replay incomplete.

66 Operational Review PARTIAL operational-review.ts; reviews route/schema. Persisted reviews/source refs exist; scheduled/CIL/Query integration unproven.

67 Experience & Institutional Knowledge PARTIAL experience.ts; foundation/institutional schemas/routes. Pathway exists; 3-event/no-contradiction promotion unproven.

68 Operational Self-Improvement PARTIAL self-improvement.ts; routes; adaptation schema. Adaptation/reset/events exist; boundary/reversibility tests absent.

69 System Economics PARTIAL system-economics.ts; routes; costs/foundation schemas. Summaries exist; CPU/memory/network/storage mostly inferred.

Additional merged follow-ups 70-73 and 76-79 are also PARTIAL under the same gate. Task 85 and 86 remain PROPOSED. Tasks 81-84 and 87 were cancelled.

Constitutional rules

1. Provenance required · PARTIAL. Provenance modules and fields exist; not every write path rejects missing provenance.
2. /internal/ never public · FAIL/PARTIAL. private-auth.ts exists, but internal routes share the /api router and broad CORS is enabled in app.ts.
3. Embeddings remain local · PARTIAL. semantic-index.ts exists; no negative leakage test.
4. No silent failures · FAIL. projector.ts uses onConflictDoNothing; restore-test is checksum-only; self-test assumes trigger installation.
5. Event Log append-only · PASS at database-trigger level, PARTIAL system-wide. ensure-append-only.mjs installs a PostgreSQL trigger; no mutation test exists.
6. Facts and Interpretations never mixed · PARTIAL. Separate ledgers exist; global write-path proof is missing.
7. Provider isolation · PARTIAL. Provider abstraction exists; direct managed OpenAI import remains in ai-providers.ts.
8. Bootstrap never reads secrets · PARTIAL. Static evidence design exists; no negative test.
9. CerbaSeal fail-closed · PASS at adapter, PARTIAL end-to-end. internal-services.ts returns HOLD on missing/unavailable/invalid service.
10. No direct CIL/CerbaSeal DB access · PASS by source inspection. Signed HTTP clients were found, not external DB imports.
11. Credentials never leak · PARTIAL. Env-based HMAC/Bearer use exists; no automated leakage test.
12. Anchors never silently contradicted · PARTIAL. Storage exists; global contradiction guard incomplete.
13. Identity before Constitution · FAIL/PARTIAL. No universal Identity · Constitution

middleware pipeline.

Critical runtime flows

Informational request: code exists across routes/ai.ts, intent.ts, query-engine.ts, context-engine.ts, context-economy.ts, model-router.ts, and explanation.ts, but the complete universal chain is not proven. CIL was unavailable and the live path falls back to managed OpenAI routing.

Consequential action: governance-engine.ts classifies unknown actions as CRITICAL and holds high-risk actions without evidence. services/internal-services.ts returns HOLD when CerbaSeal is unavailable. No integration test proves every consequential execution route uses this boundary before execution.

Project Bootstrap: project-bootstrap.ts reads static evidence and creates draft facts, interpretations, graph material, timeline information, and questions. Full repository-connected execution and secret-exclusion tests are absent.

Event replay: foundation-events.ts provides replayAggregate and projector.ts provides limited universal-object replay. It is not full-system reconstruction, has inconsistent cursor semantics, mutates live projections, lacks reset/dry-run/transaction options, and can silently skip conflicts.

Fact versus Interpretation: storage and routes are genuinely separate; all creation paths are not tested.

CIL: T1/T2/T3 contracts, HMAC, cost, provenance, reuse, escalation, drift, and contradiction fields are represented. Real tier execution and frontier reduction were not demonstrated.

Institutional Knowledge: Experience, Lesson, Pattern, and Institutional Knowledge records exist. The required independent-confirmation/no-contradiction gate is not proven.

Self-Improvement: adaptation categories, metrics, reset/disable concepts, and events exist. Boundary protection and reversibility are not integration-tested.

Runtime wiring map

Console and Android call /api/* through Express app.ts. app.ts applies privateAuth() and recoveryModeGuard, then mounts the router. index.ts starts recovery boot, scheduled jobs, provider registration, internal service registration, event subscriptions, Android push delivery, and Executive Loop scheduling. CIL and CerbaSeal are called over signed HTTP; LEE does not access their databases.

Database ownership map

Foundation tables own universal objects, Event Log, facts, interpretations, provenance, identity, Constitution, Brain Versions, anchors, assumptions, decisions, institutional knowledge, experiences, lessons, graph nodes/edges, timelines, and milestones. Operational schemas own objectives, momentum, opportunities, capacity, initiatives, operational intelligence, patterns, reviews, adaptations, economics, readiness, allocation, recovery, scheduler, provider state, and backups.

The schemas are extensive, but many engines import @workspace/db directly rather than retrieving through Query Engine.

Event producer/consumer map

Producers include foundation events, governance, reviews, experience, self-improvement, economics, bootstrap, backups, self-test, scheduler, and operational engines. Consumers include in-process subscribers, Android push, Executive Loop, operational memory, self-improvement, and the limited projector. There is no durable subscriber cursor, retry ledger, or restart

redelivery mechanism.

API surface map

Major groups include `/api/health`, `/api/ai/*`, `/api/ledgers/*`, `/api/query/*`, `/api/explanations/*`, `/api/provenance/*`, `/api/governance/*`, `/api/internal-services/*`, `/api/bootstrap/*`, `/api/providers/*`, `/api/backups/*`, `/api/brain-versions/*`, `/api/reviews/*`, `/api/institutional-knowledge/*`, `/api/self-improvement/*`, `/api/economics/*`, `/api/execution-readiness/*`, `/api/resource-allocation/*`, `/api/portfolio-dependency/*`, `/api/android/*`, `/api/recovery/*`, and `/api/self-test/*`.

Live checks returned HTTP 200 for `/api/health`, `/api/internal-services/health`, `/api/backups/status`, and `/api/execution-readiness`. Intuitive paths `/api/manifest`, `/api/self-test/history`, `/api/portfolio-dependency`, `/api/operational-intelligence`, and `/api/system-economics` returned 404, demonstrating route naming/smoke coverage drift.

Persistence audit

Projects, people, facts, interpretations, assumptions, anchors, decision heuristics, institutional knowledge, objectives, simulations, opportunities, reviews, operational patterns, momentum, readiness, provider state, policies, identity, Brain Versions, and Event Log have database structures. CIL call records and CerbaSeal references are partial. Table existence does not prove correct lifecycle persistence or event reconstruction.

Backup audit

Backup creation and payload checksums exist in `routes/backups.ts` and `schema/backups.ts`. Brain Version metadata is included and the manifest lists Event Log/ledger tables. There is no proven System Manifest export, complete database backup, isolated restore, foreign-key validation against restored state, Event Log rebuild, or restored-state comparison. POST `/backups/:id/test-restore` only recomputes the payload checksum.

Self-Test audit

`self-test.ts` persists reports and runs Engine, API, Policy, Domain Events, Context Economy, Backup/Event Log, and Provider suites. The API suite tests registration, Policy tests active rows, Events uses a synthetic event, append-only assumes trigger installation, and Backup checks only for a latest backup row. No repository test files or test script cover replay, restore, recovery, concurrency, or runtime smoke behavior.

UI audit

Console source is `App.tsx` plus page components. Real rendered routes include schedule, projects, portfolio, knowledge map, observations, strategy, anchors, simulations, reflections, learning, people, workspace, Constitution, confidence, evidence, assumptions, impact, timeline, explanation, policies, and Android pairing. These generally use live fetch calls and some loading/error states. Navigation also lists costs, governance, backups, objectives, organization, knowledge, institutional knowledge, self-improvement, economics, events, reviews, and health, but those are not all represented in the central route conditional.

Android source is under `app`, `LeeContext.tsx`, `storage.ts`, and `api.ts`. It includes local capture, pairing, Ask Lee, alerts, approvals, waiting, uncertainty notices, and offline persistence. Device-level restart, conflict, and full governed-write tests are absent.

Coverage percentages

Measure Coverage

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Task names represented by source modules/tables/routes approximately 100%

Tasks with persisted database structures approximately 90-95%
Tasks with API exposure approximately 90%
Tasks with UI exposure approximately 60-70%
Tasks with meaningful failure behavior approximately 45-55%
Tasks with real automated behavioral tests effectively 0%
Tasks satisfying all eight FULLY IMPLEMENTED requirements 0%

Severity summary

P0 · Architecture-breaking

1. Full event sourcing is not implemented; routes commonly write projections directly.
2. Backup restore is not real; restore-test is checksum validation only.
3. CIL is not operationally available.
4. CerbaSeal end-to-end execution blocking is not proven.
5. Identity · Constitution ordering is not universally enforced.

P1 · Major incomplete functionality

Query Engine is not universal; Self-Test tests metadata more than behavior; Institutional Knowledge promotion is unproven; Self-Improvement boundaries are not tested; provider isolation is convention-based; internal authorization is not comprehensively proven; Console navigation has route drift; live provider sync/execution is unproven; Operational Intelligence and Bootstrap lack end-to-end evidence.

P2 · Wiring/persistence/test weaknesses

No repository test suite; in-process-only event subscribers; inconsistent replay cursor; no replay dry-run/reset; trigger not inspected by self-test; economics metrics mostly inferred; no credential leakage test; provenance/ownership not universal; UI fetch failures can become empty state; no checked-in migration/restore artifact.

P3 · Polish

Route naming is not self-documenting; navigation drift; Manual CSS @import warning; large API bundle warning.

Direct answers

1. Is Project LEE implemented according to Version 12? No; it is a broad partial implementation.
2. Which rules are enforced? Database append-only protection, CerbaSeal adapter fail-closed behavior, separate external HTTP boundary, separate Fact/Interpretation ledgers, unknown-action critical classification, and Android token security are strongest. Many others are partial/documented.
3. Are CIL and CerbaSeal integrated? Signed HTTP contracts/clients are real; live service integration is not proven. CIL unavailable; CerbaSeal returned 404.
4. Does CerbaSeal fail closed? Yes at the adapter boundary; not proven across every consequential path.
5. Does CIL reduce frontier dependence? Not demonstrated; live routing fell back to managed OpenAI.
6. Is Event Log true event sourcing? No; mainly append-only audit/activity logging with limited projection replay.
7. Can LEE reconstruct state from events? Only narrowly, not complete system state.
8. Is Fact/Interpretation separation real? Yes in storage; partial as universal enforcement.
9. Is Query Engine universal? No; many engines query DB directly.
10. Is Bootstrap functional? Partially; static evidence-first code exists, but full runtime/security proof absent.
11. Is Operational Intelligence genuinely calculating priorities? Partially; ranking/scheduling exist, complete evidence-driven output unproven.
12. Is Institutional Knowledge earned from repeated outcomes? Pathway exists; promotion gate

unproven.

13. Is Self-Improvement real and bounded? Partially; adaptations/boundaries represented, not integration-tested.

14. Is Backup + Test Restore functional? Backup/checksum yes; real restore no.

15. Can this build be trusted for persistent internal use? Only as a development/prototyping system with explicit limitations; not yet as a recovery-safe operating system for consequential workflows.

Final verdict

Project LEE has real foundations, substantial persistence, a meaningful domain model, running workflows, a local-first Android companion, and a genuine CerbaSeal fail-closed adapter. It does not yet satisfy Version 12 as a fully implemented, independently verifiable, event-reconstructable founder operating system.